

Disproof of conjecture 00000008371

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Verdict: FALSE

We disprove conjecture 00000008371 as stated. The conjecture is a *conjunction* of four laws about “space trees”. A conjunction is false as soon as one conjunct is false, and here at least one conjunct fails for every admissible n : the asserted counting formula is not even an integer at $n = 3$, and it disagrees with the asserted Johnson-graph vertex count for $n \geq 3$. The refutation is elementary and unconditional; the Speyer–Sturmfels count $(2n - 5)!!$ is cited only as context.

1 The conjecture

Definition 1 (as stated). The *Nielsen cells* of the tropical Grassmannian are the cells of the tropicalization of $\text{Gr}(2, n)$ (the “space trees”); let `space_tree` denote their number.

Conjecture. `space_tree` (the count of space trees) is the type count $(n - 2)!(n - 3)!/2$ (a type counting law); the mod-2 pattern of the count (parity) follows the power law 2^{n-3} (a power parity law); the adjacency graph of cells (shared faces of adjacent cells) is the Johnson graph $J(n, 2)$ (a Johnson adjacency law); and the complete spectrum of the adjacency graph (eigenvalues) is a difference set (a difference set spectrum law).

Thus the conjecture asserts the conjunction

$$\underbrace{\text{space_tree} = \frac{(n-2)!(n-3)!}{2}}_{\text{(A) type count}} \wedge \underbrace{\text{parity} = 2^{n-3}}_{\text{(B) power parity}} \wedge \underbrace{\Gamma \cong J(n, 2)}_{\text{(C) Johnson adjacency}} \wedge \underbrace{\text{spec}(\Gamma) \text{ is a difference set}}_{\text{(D) spectrum}}.$$

We write

$$F(n) := \frac{(n-2)!(n-3)!}{2}, \quad J_n := \text{number of vertices of } J(n, 2) = \binom{n}{2} = \frac{n(n-1)}{2}.$$

2 Failure 1: the formula is not a count at $n = 3$

Lemma 2. $F(3) = \frac{1!0!}{2} = \frac{1}{2}$.

Proof. $3 - 2 = 1$ and $3 - 3 = 0$, so $(3 - 2)! = 1! = 1$ and $(3 - 3)! = 0! = 1$. Hence $F(3) = 1 \cdot 1/2 = 1/2$. $\square$

Theorem 3. $F(3)$ is not a non-negative integer. Consequently conjunct (A) fails at $n = 3$: a cardinality (a count of cells) cannot be $\frac{1}{2}$.

Proof. $1/2 \notin \mathbb{Z}_{\geq 0}$. The quantity space_tree is by definition the number of cells of a cell complex, hence a non-negative integer. The conjecture identifies it with $F(3) = 1/2$, a contradiction. $\square$

Remark 4. The same objection makes conjunct (B) ill-defined at $n = 3$: a half-integer has no parity. This is a secondary observation; Failure 1 already refutes the conjunction on its own.

The relevant values of F are

$$F(3) = \frac{1}{2}, \quad F(4) = 1, \quad F(5) = 6, \quad F(6) = 72.$$

3 Failure 2: the formula never equals the Johnson vertex count

The Johnson graph $J(n, 2)$ has as vertices the 2-subsets of an n -set, so it has exactly $\binom{n}{2} = n(n-1)/2$ vertices:

$$J_3 = 3, \quad J_4 = 6, \quad J_5 = 10, \quad J_6 = 15, \quad J_7 = 21, \quad J_8 = 28.$$

Conjunct (A) and conjunct (C) are mutually inconsistent.

Theorem 5. $F(n) \neq \binom{n}{2}$ for every integer $n \geq 3$. Hence for every $n \geq 3$ at least one of the conjuncts (A), (C) is false.

Proof. Split into three ranges.

$n = 3, 4, 5$: direct evaluation gives

$$F(3) = \frac{1}{2} \neq 3 = \binom{3}{2}, \quad F(4) = 1 \neq 6 = \binom{4}{2}, \quad F(5) = 6 \neq 10 = \binom{5}{2}.$$

$n = 6$: $F(6) = 72 \neq 15 = \binom{6}{2}$.

$n \geq 6$ (all remaining n): we show $F(n) > \binom{n}{2}$ by induction on n . The base case $n = 6$ is $72 > 15$. For the step, note the recurrence

$$F(n+1) = \frac{(n-1)!(n-2)!}{2} = (n-1)(n-2) \cdot \frac{(n-2)!(n-3)!}{2} = (n-1)(n-2)F(n).$$

Assume $F(n) > \binom{n}{2}$. Since $(n-1)(n-2) > 0$,

$$F(n+1) = (n-1)(n-2)F(n) > (n-1)(n-2)\binom{n}{2} = (n-1)(n-2)\frac{n(n-1)}{2} = \frac{n(n-1)^2(n-2)}{2}.$$

For $n \geq 6$ we have $(n-1)(n-2) \geq 5 \cdot 4 = 20$, hence

$$F(n+1) > (n-1)(n-2)\binom{n}{2} \geq 20\binom{n}{2} = 10n(n-1) > \frac{n(n+1)}{2} = \binom{n+1}{2},$$

because $10n(n-1) > n(n+1)/2$ for $n \geq 6$. This closes the induction. Therefore $F(n) > \binom{n}{2}$, in particular $F(n) \neq \binom{n}{2}$, for every $n \geq 6$. $\square$

Remark 6. The theorem shows that (A) and (C) cannot both hold: if (A) held, the number of vertices of the adjacency graph would be $F(n)$, but (C) says it is $\binom{n}{2}$, and these differ for all $n \geq 3$. This argument is independent of the $n = 3$ integrality issue.

4 Failure 3: comparison with the known count (context)

It is a theorem of Speyer and Sturmfels [1] that the tropical Grassmannian $\text{Trop}(\text{Gr}(2, n))$ has exactly

$$(2n - 5)!! = 1 \cdot 3 \cdot 5 \cdots (2n - 5)$$

maximal cones; equivalently, the space of phylogenetic trees with n leaves has $(2n - 5)!!$ vertices / combinatorial types. These are the “space trees” the conjecture is about. The values are

$$1, 3, 15, 105, 945 \quad (n = 3, 4, 5, 6, 7),$$

which must be compared with the conjecture’s own assertions:

n	$F(n) = (n - 2)!(n - 3)!/2$	$J_n = \binom{n}{2}$	$(2n - 5)!!$ (known)
3	1/2	3	1
4	1	6	3
5	6	10	15
6	72	15	105
7	1440	21	945
8	43200	28	10395

Neither $F(n)$ nor $\binom{n}{2}$ agrees with the known count $(2n - 5)!!$ for any $n \geq 3$. In particular the conjectured formula does not count the objects it names. This section is context only: Failures 1 and 2 already refute the conjunction without any input from tropical geometry.

5 The $n = 3$ caveat, and why the refutation survives it

For $n = 3$ the Grassmannian $\text{Gr}(2, 3) \cong \mathbb{P}^2$ is a single cell and $\text{Trop}(\text{Gr}(2, 3))$ is a single point, so $(2n - 5)!! = 1$. If the author intended the conjecture only for $n \geq 4$, then the “ $\frac{1}{2}$ ” of Failure 1 does not occur (the formula would be evaluated only where it is an integer), and we say so honestly: *Failure 1 alone is sensitive to the range*. We therefore do not rest the disproof on it.

The disproof is unaffected, because the other failures do not use $n = 3$:

- **Failure 2** holds at $n = 4, 5, 6$: $F(4) = 1 \neq 6 = \binom{4}{2}$, $F(5) = 6 \neq 10 = \binom{5}{2}$, $F(6) = 72 \neq 15 = \binom{6}{2}$; and Theorem 5 gives $F(n) > \binom{n}{2}$ for all $n \geq 6$. So for every $n \geq 4$ the conjuncts (A) and (C) are mutually inconsistent.
- **Failure 3** holds at $n = 4, 5, 6, 7$: $1 \neq 3$, $6 \neq 15$, $72 \neq 105$, $1440 \neq 945$, and F grows far faster than $(2n - 5)!!$ super-exponentially, so agreement for larger n is impossible.

Thus, whether or not one excludes $n = 3$, the conjunction is false for every admissible n : excluding $n = 3$ removes Failure 1 but leaves Failures 2 and 3.

6 Conclusion

The conjecture asserts the conjunction $(A) \wedge (B) \wedge (C) \wedge (D)$. At $n = 3$, conjunct (A) evaluates to $\frac{1}{2}$, which is not a count (Failure 1); for every $n \geq 4$, conjunct (A) contradicts conjunct (C), whose vertex count is $\binom{n}{2}$ (Failure 2); and the formula disagrees with the established Speyer–Sturmfels count $(2n - 5)!!$ (Failure 3). Every conjunct is a necessary part of the conjecture, so the conjecture is **FALSE**.

References

- [1] D. Speyer and B. Sturmfels, *The tropical Grassmannian*, Advances in Geometry **4** (2004), no. 3, 389–411.